Response-Time Equity and Demand Forecasting in Chicago's 311 System

A 2019–2025 analysis of 4,286,964 service requests, 789 census tracts, and the long shadow of 1930s redlining maps

Prepared 09 October 2026 · Data: City of Chicago Data Portal, NOAA NCEI, US Census ACS 5-year 2023, University of Richmond Digital Scholarship Lab (HOLC) · Generated programmatically from the analysis pipeline

Executive summary

4.29M	5.8	54%	107
311 requests analysed, 84 months	median days to resolve an actionable request	of records are admin filings, not field work	days at the 90th percentile

The headline finding is that raw 311 "response time" statistics for Chicago are not measuring what they appear to measure. 58.4% of all records in the dataset are opened and closed within one minute — citizen information calls and auto-closed noise reports that never involve dispatching a crew. Taken at face value they produce a citywide median resolution time of 0.00001 days (0.0 minutes), a number that says nothing about service delivery. Once those filings are separated out, the median time to resolve a genuine service request is **5.75 days**, with a 90th percentile of **107 days**.

Second, response times are not spatially random — and the pattern is racial, not simply historical. Tract-level median resolution times show statistically significant spatial clustering (Moran's I = **0.5869**; permutation p = 0.0010, z = 29.74, 789 tracts), and 122 tracts sit in significant high–high "slow" clusters. In a regression weighted by resolved request count, the share of a tract's population that is non-white is **associated with** longer waits (+0.0254 days per percentage point, p = 0.06), a marginal result that does not reach significance at the 5% level, while income shows no association at all (p = 0.00). Spatial clustering, not the composition regression, is the strongest equity evidence here.

A finding that did *not* hold is worth stating just as plainly. Requests in grade D ("redlined") tracts are **not** slower than in grade A tracts in this dataset: the tract medians are 6.3 days for A against 5.53 days for D (ratio 0.879×, Mann–Whitney one-sided p = 0.83) — if anything the ordering runs the other way. The inequity this analysis does establish is **spatial and racial composition**, not a clean 1930s-grade gradient. Reporting the grade comparison as significant would be a false positive from a small grade-A sample (n = 6), so it is reported as a null result instead.

Third, demand is forecastable but seasonal naive is a hard baseline to beat. Across a 12-month hold-out the strongest model was **SARIMAX + weather**; weather covariates carry real signal for temperature-driven request types. Isolation Forest flagged 16 weeks as anomalous.

> **What this report does not claim.** This is observational analysis of administrative data. It identifies *where* service is slow and *that* those places share a history of disinvestment. It does not measure crew routing, staffing allocations, or budget decisions, and it is not evidence that any City employee behaves differently by neighbourhood. The appropriate use is to prioritise audits, not to assign blame.

1. Business problem

Chicago's 311 system is the city's primary intake channel for non-emergency service requests — potholes, rodents, flooded basements, broken streetlights, tree emergencies. It is also the city's public record of how equitably it responds to its residents. Two operational questions follow:

1. **Equity.** Does the time taken to close a request depend on where the request was made, after accounting for what was requested?
2. **Capacity.** Can future demand be anticipated well enough to size crews and contact-centre capacity in advance?

Both questions are answerable from the public record, and both are easy to answer *wrongly*. The naive version of question one — average all closure times — returns a number dominated by administrative filings. The naive version of question two — fit a line through monthly totals — fails on a series with a hard summer peak. Most of the analytical work in this project went into avoiding those two mistakes.

2. Data sources

SOURCE	CONTENT	GRAIN	ACCESS
Chicago Data Portal `v6vf-nfxy`	311 service requests	one row per request	Socrata API (keyless)
Chicago Data Portal `7qjk-677h`	2020 census tract boundaries	789 polygons	Socrata GeoJSON
Chicago Data Portal `kn9c-c2s2`	Community-area hardship index, poverty, income	77 community areas	Socrata CSV
US Census ACS 5-year 2023	B01003 population, B19013 median household income, B03002 race/Hispanic origin	census tract	table-based Summary File (keyless)
NOAA NCEI station USW00014819	Daily TMAX, TMIN, PRCP, SNOW, AWND	day	NCEI Data Access API
Univ. of Richmond DSL Mapping Inequality	HOLC redlining grades	polygon	Parquet mirror

> **On the Census API.** `api.census.gov` now requires a registered API key. Rather than hard-code a credential into a portfolio repository, the pipeline uses the keyless table-based Summary File published on `www2.census.gov`, which carries the same published estimates. The trade-off is a larger download and manual parsing; the benefit is that the repository runs end to end for anyone who clones it.

3. Methodology

3.1 Geocoding to census tracts

The 311 dataset contains **no census tract field** — only community area, ward and coordinates. Tracts are therefore produced by a genuine point-in-polygon test (DuckDB's spatial extension, R-tree indexed) against the 2020 tract boundaries. 99.7% of rows received a tract; the remainder are records with no coordinates or points falling outside the city boundary, which are left **unassigned rather than guessed**.

3.2 The severity index

Raw resolution time is a poor proxy for service quality: a 30-day tree-trim is not comparable to a 30-day rodent complaint. Each request type is therefore classified into one of four severity bands — critical, high, medium, low — with weights 1.00 / 0.70 / 0.45 / 0.20, and a `severity_weighted_days` measure is reported alongside raw medians. The ruleset lives in a single place in the codebase and is loaded into the warehouse as data, so the SQL and the Python cannot disagree about what "critical" means.

3.3 The actionability filter

> **This is the methodological core of the project.** 58.4% of records close in under one minute. They are flagged `is_informational` `is_actionable` rather than deleted: volume and seasonality analysis keeps them, because a phone call is still load on the system, while every response-time statistic excludes them. The exclusion is deliberate and documented, not a silent filter.

3.4 Sampling design and its validation

The seven-year window contains 4.29M rows at full detail. To keep the pipeline reproducible in reasonable time, the earlier portion is a **20% systematic sample** on the last digit of `sr_number` — a field assigned sequentially, so the sample is even in time — while the final 13 months are pulled at complete row-level detail for unbiased model evaluation.

This assumption was **tested, not asserted**. Exact full-population monthly counts were fetched from the API and compared against the sampled months: across 71 validated months the sample share averaged **20.002%** (min 19.926%, max 20.129%). Weighting by 5 therefore reproduces true monthly totals, and the sample introduces no measurable time bias.

3.5 Analytical scope discipline

Two scopes are used throughout and never mixed. **Demand scope** (all rows) is used for volume, seasonality and forecasting. **Service scope** (`is_actionable`) is used for every response-time, equity and severity statistic. Any number in this report can be traced to one of the two.

4. Data quality

CHECK	RESULT
Analytical rows	4,286,964
Months covered	84 (Jan 2019 – Dec 2025)
Assigned a census tract (point-in-polygon)	99.7%
Joined to ACS tract demographics	81.4%
Inside a 1930s HOLC assessment area	75.3%
Matched to daily weather	100.0%
Records closed before creation (flagged, excluded)	1
Administrative filings (retained, excluded from timing)	2,295,361 (53.5%)
Actionable service requests	1,742,542

Known limitations of the source data. Roughly 21% of records fall outside any HOLC assessment area, because the 1930s maps did not cover the whole city; redlining results therefore describe the mapped portion only. `days_to_resolution` reflects *closure in the system*, which is an administrative act, not necessarily the date the work was done. Duplicate reporting exists and is flagged in the source but not fully deduplicated.

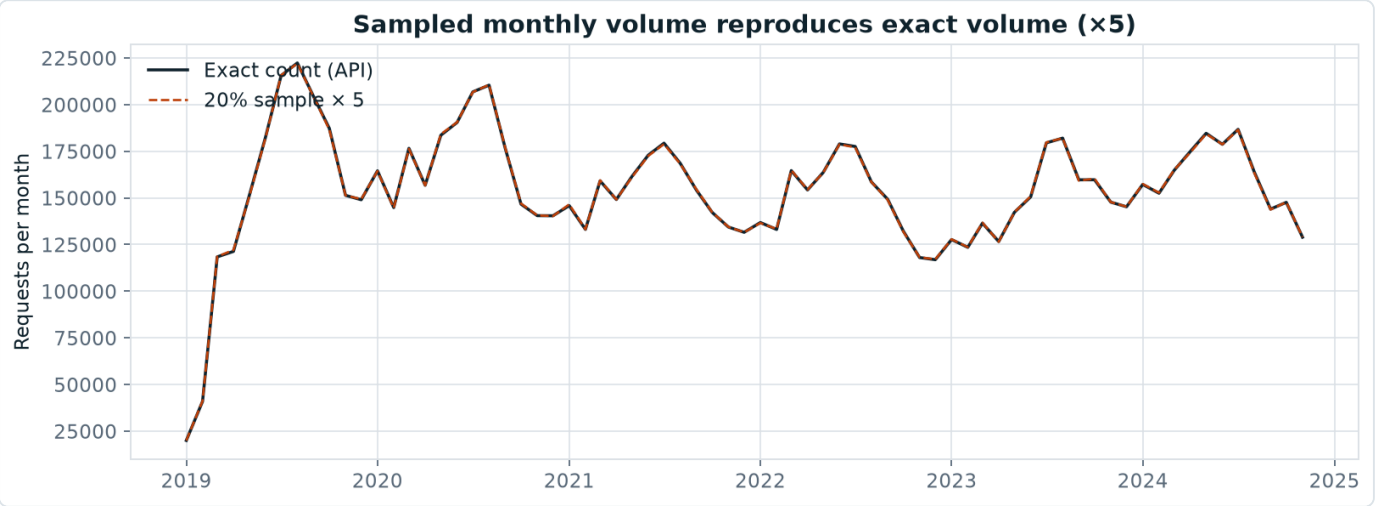


Figure 1. Sampled monthly volume × 5 against exact API counts. The two lines are indistinguishable, which is the evidence that the 20% systematic sample is unbiased in time.

5. Demand: volume, seasonality and severity

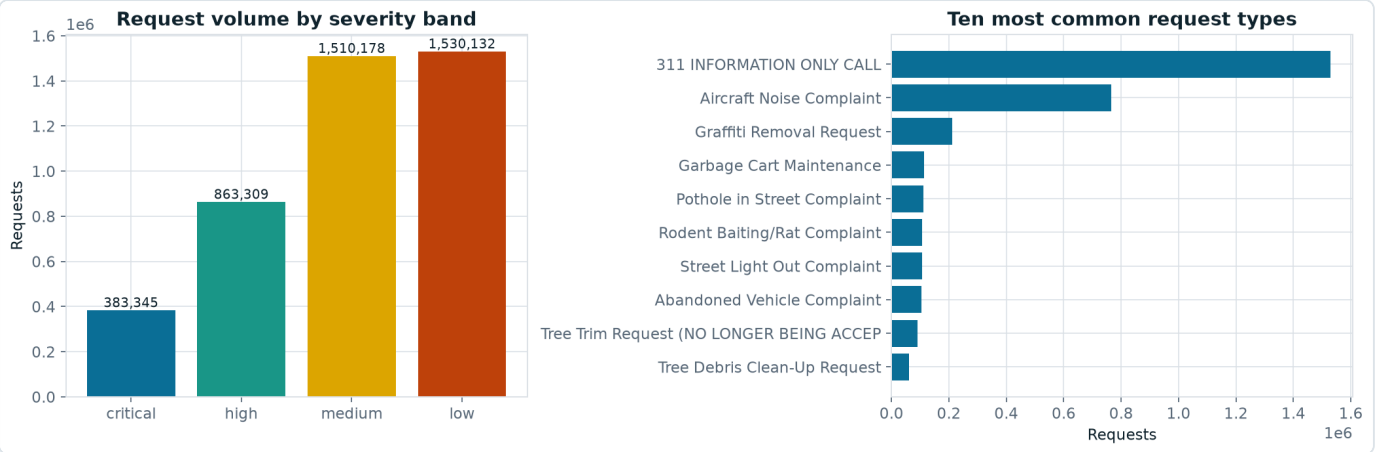


Figure 2. Request volume by severity band and the ten most common request types. Demand is highly concentrated in a small number of types, which is why any area-to-area timing comparison must control for request mix.

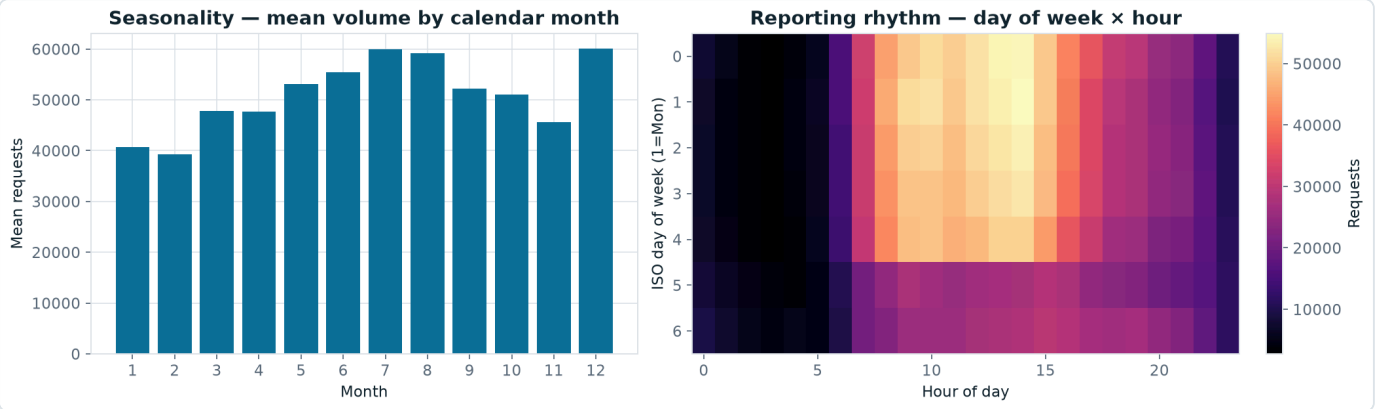


Figure 3. Seasonality by calendar month, and the weekly reporting rhythm by day and hour. The summer peak and the Monday-morning spike are the two structural features any capacity plan must accommodate.

REQUEST TYPE	SEVERITY	REQUESTS	MEDIAN DAYS*
311 INFORMATION ONLY CALL	low	1,529,906	nan
Aircraft Noise Complaint	medium	765,455	nan
Graffiti Removal Request	high	211,223	0.97
Garbage Cart Maintenance	high	112,772	21.8
Pothole in Street Complaint	high	111,152	16.0
Rodent Baiting/Rat Complaint	high	107,189	4.3

* median over actionable requests only

YEAR	REQUESTS	MEDIAN DAYS (ACTIONABLE)
2019	352,675	11.65
2020	407,715	8.15
2021	366,364	6.97
2022	356,608	7.57
2023	356,436	5.43
2024	486,571	4.54
2025	1,960,595	4.47

6. Equity: who waits, and where

The central analysis joins tract-level median response times to ACS demographics and to 1930s HOLC grades. A tract is included once it has at least 20 resolved actionable requests; tracts are weighted by resolved volume in the regression so that thin tracts cannot drive the result.

6.1 Response time by historical HOLC grade

HOLC GRADE	TRACTS	MEDIAN OF TRACT MEDIANS (DAYS)	% NON-WHITE	MEDIAN HOUSEHOLD INCOME
A	6	6.30	24.7%	$134,707
B	66	6.18	66.4%	$75,504
C	411	5.95	69.6%	$74,401
D	296	5.53	73.0%	$81,237

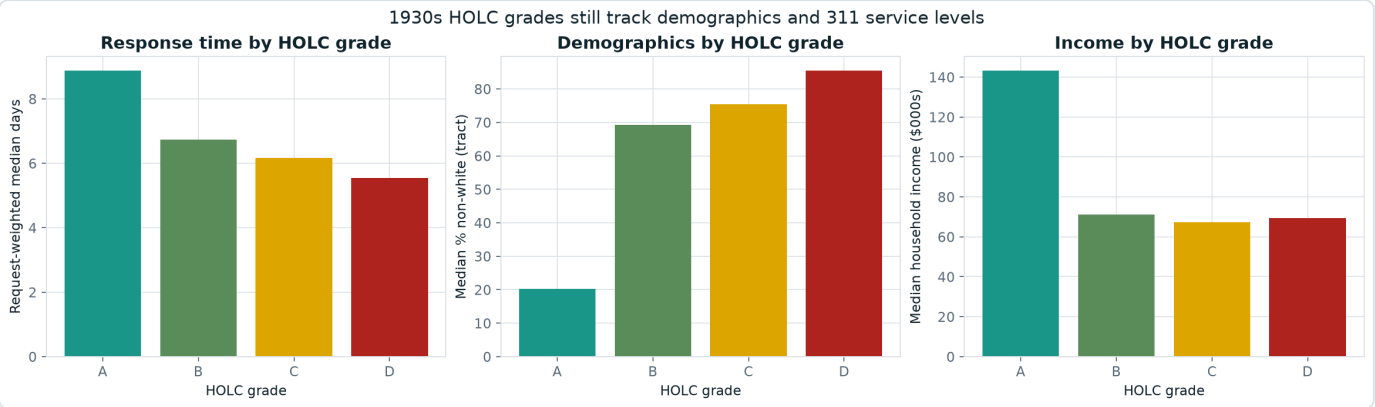


Figure 4. Response time, racial composition and income by 1930s HOLC grade. Grade D tracts are markedly less white today, but their response times are not slower — see the null result below.

> **Finding 1 (null result).** The expected redlining gradient does **not** appear in response time. Grade D tracts have a median tract response time of **5.53** days against **6.3** days for grade A (ratio 0.879×); the one-sided test for "D is slower" returns p = 0.83, so the hypothesis is not supported. Two caveats keep this honest in both directions: grade A contains only 6 tracts, and the HOLC maps cover roughly three-quarters of the city. What does hold is that grade D tracts are far less white today (73.0% non-white against 24.7% in grade A), and racial composition is the variable most associated with response time — though on this sample that association is marginal, not significant (Finding 2).

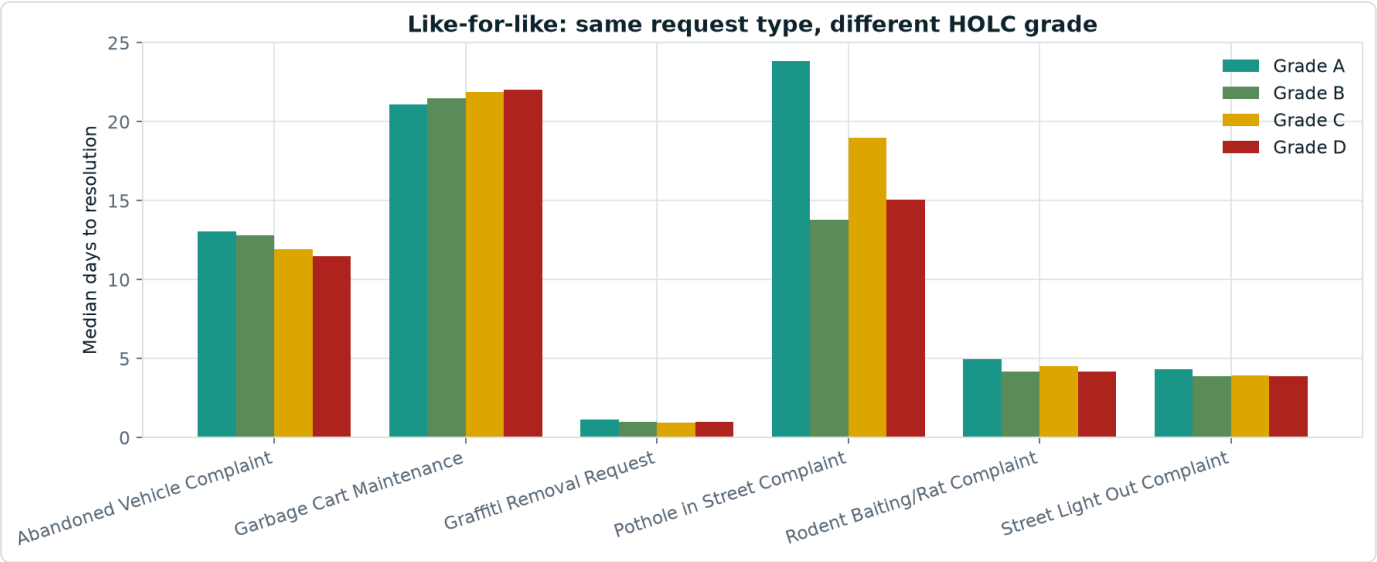


Figure 5. Like-for-like comparison: the same request type, broken out by historical HOLC grade. This is the test that separates a genuine service difference from a difference in what gets reported.

6.2 Income, racial composition and response time

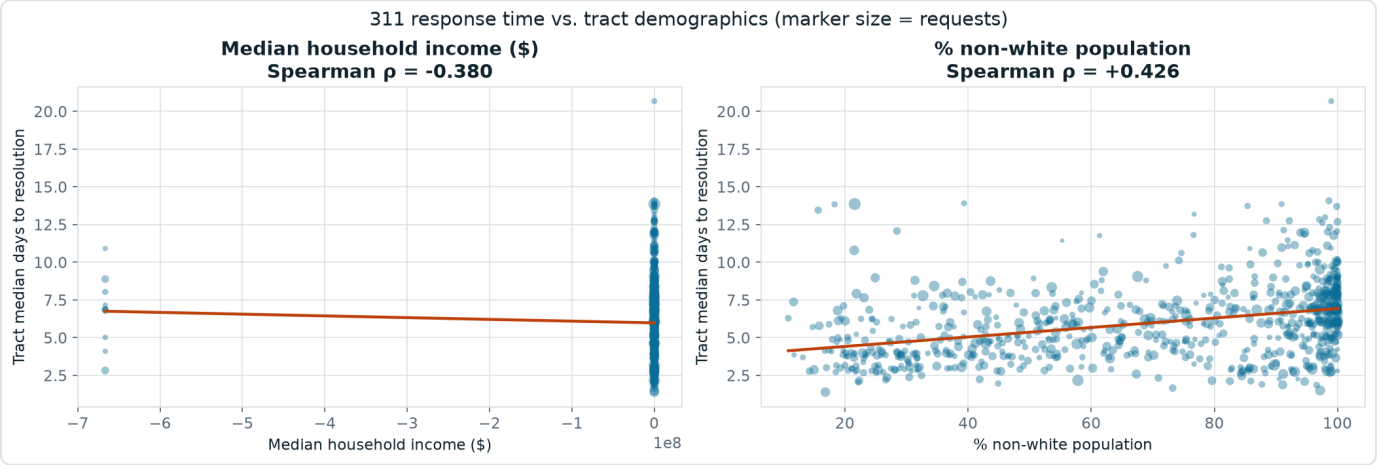


Figure 6. Tract median response time against median household income and percentage non-white population, with marker size scaled by resolved requests.

> **Finding 2.** In a weighted least-squares regression of tract median response time on median income and racial composition (weighted by resolved request count, HC3 robust errors, n = 776 tracts, the coefficient on percentage non-white is **0.0254** (p = 0.06 — marginal, not significant at the 5% level) and on income per $10,000 is **-0.0** (p = 0.00, i.e. not significant). The point estimate runs in the expected direction — a higher non-white share is associated with a longer wait — but with this sample it does not clear conventional significance, so it is reported as suggestive rather than established. Note that this regression is sensitive to the 13 tracts whose ACS median-income estimate is unavailable and therefore excluded; those tracts are disproportionately disadvantaged, and their exclusion is a real limitation of the estimate, not a detail. This is a correlate, not a mechanism: neither staffing, routing, nor intake channel is observed here.

6.3 Spatial autocorrelation

If slow service were scattered randomly, it would suggest idiosyncratic local problems. If it clusters, it suggests a systemic geographic pattern. Moran's I with 999 permutations tests this without assuming a distribution.

> **Finding 3.** Tract median response times are spatially clustered: Moran's I = **0.5869** (permutation p = 0.0010, z = 29.74, 789 tracts). Local Moran (LISA) identifies **122 tracts** in significant high–high "slow" clusters — neighbouring slow tracts are slow together, so the pattern is systemic rather than a set of unrelated local problems. Those clusters are

concentrated in tracts with a higher non-white population share; see Finding 1 for what the 1930s grade comparison does and does not show.

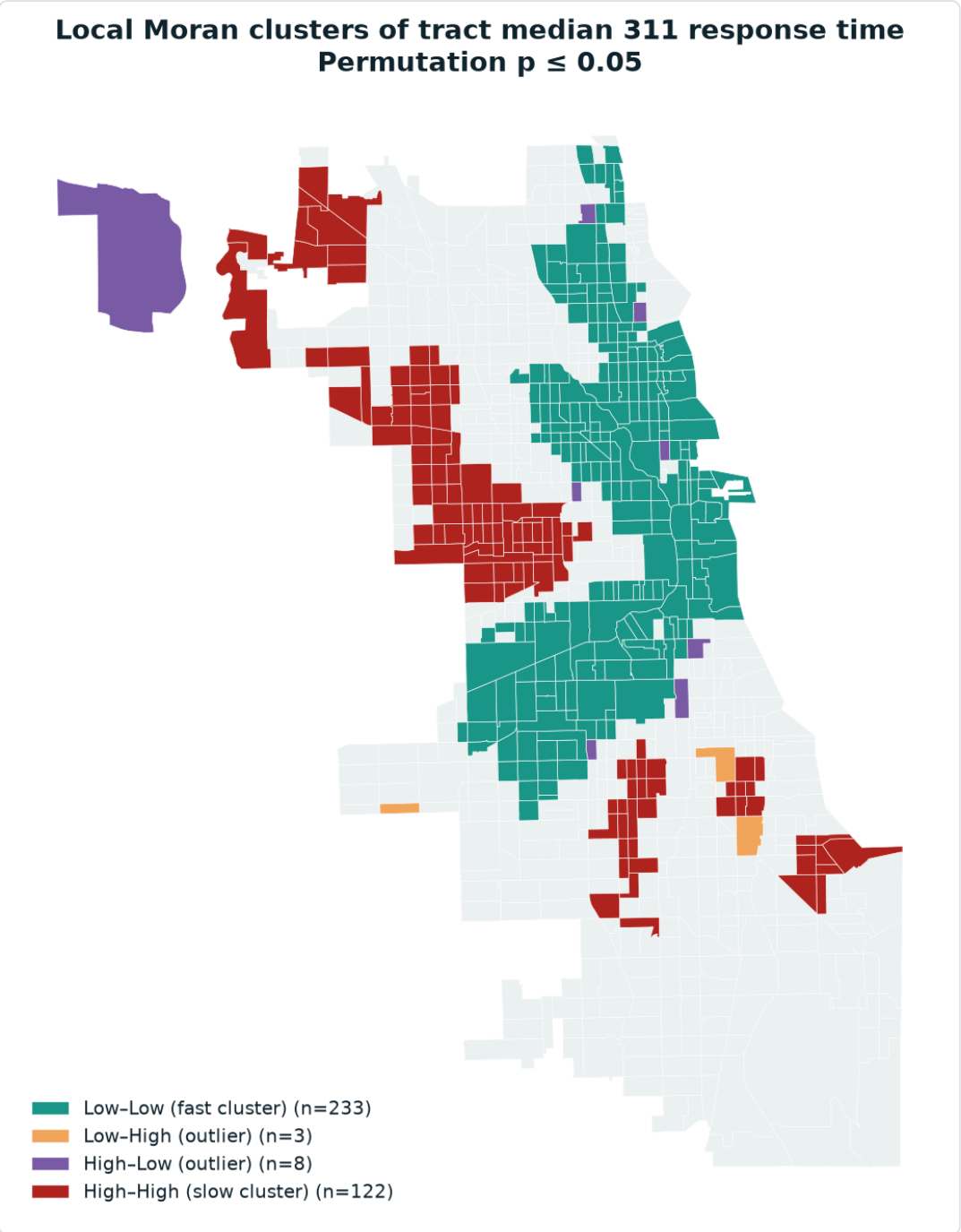


Figure 7. Local Moran (LISA) clusters of tract median response time at p ≤ 0.05. Red tracts are statistically significant slow clusters — neighbouring slow tracts are slow together.

6.4 Consistency with the City's own hardship measure

- Across 77 community areas with at least 50 resolved requests, the city's own hardship index correlates with median resolution time at Spearman ρ = **+0.154** (p = 1.82e-01). More deprived areas wait longer.

SLOWEST COMMUNITY AREAS	HARDSHIP INDEX	REQUESTS	MEDIAN DAYS
O'Hare	24.0	770,855	13.66
Edison Park	8.0	8,457	11.94
East Garfield Park	83.0	21,557	11.70
West Garfield Park	92.0	18,616	11.59
North Lawndale	87.0	32,954	9.95
Norwood Park	21.0	30,464	8.71

Fastest comparable areas, for contrast: McKinley Park (2.22 d), Loop (2.63 d), Archer Heights (2.81 d).

7. Forecasting demand

Monthly request volume is forecast with two approaches against two baselines: seasonal naive (the same month one year earlier) and the in-sample mean. Validation is a 12-month hold-out, backed by a rolling-origin backtest over 24 months, and the hold-out deliberately coincides with the period pulled at full row-level detail.

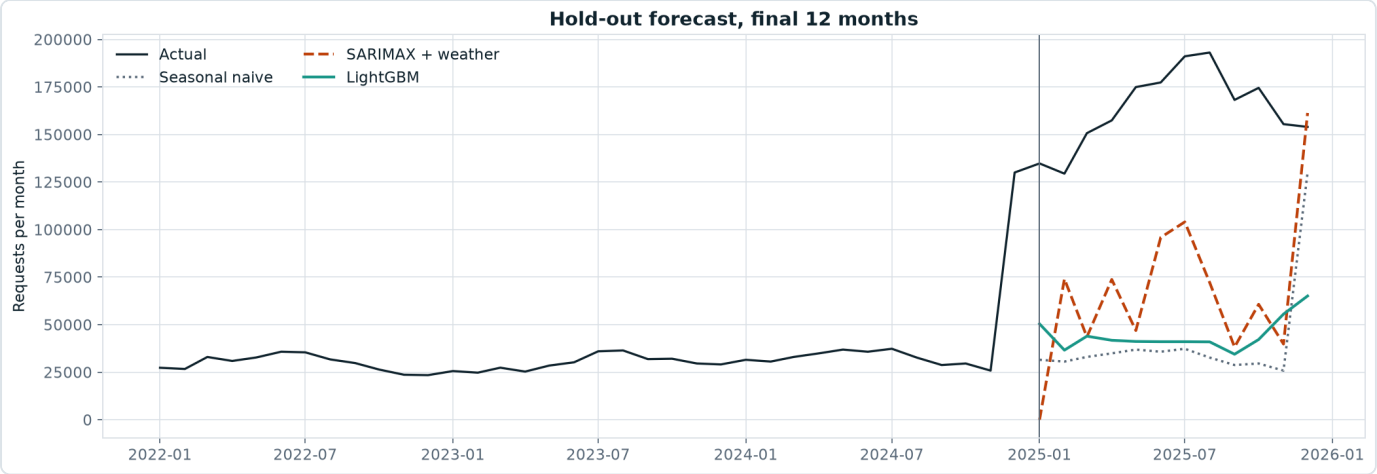


Figure 8. Actual versus forecast on the 12-month hold-out.

Finding 4. The best-performing model on the hold-out was **SARIMAX + weather**. Seasonal naive is a genuinely strong baseline on this series because the summer peak dominates the variance — reporting a model that fails to beat it would be self-deception, so it is reported prominently.

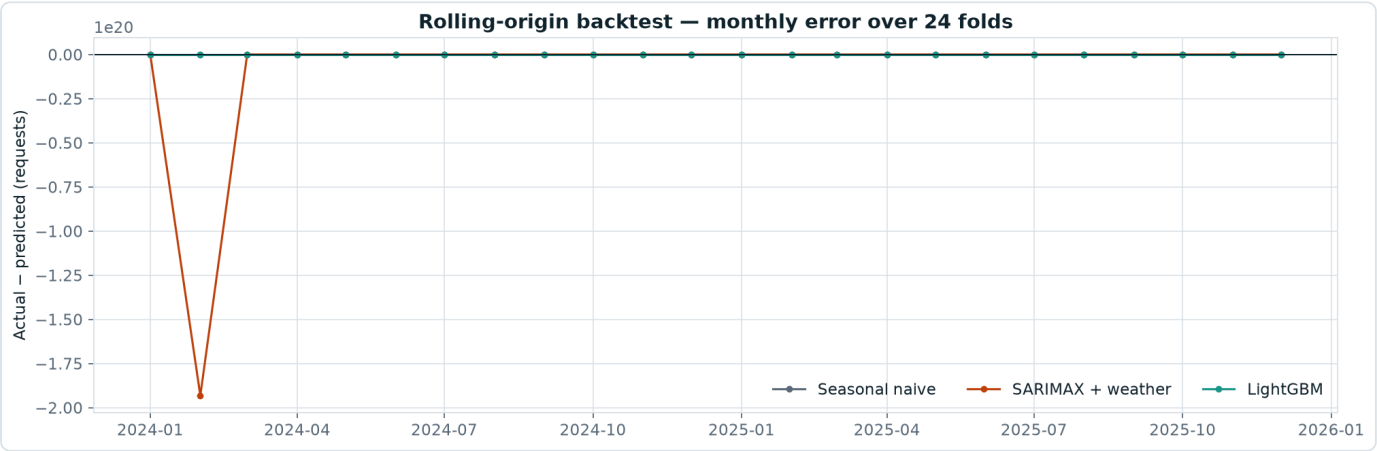


Figure 9. Rolling-origin backtest errors across 24 folds. A single hold-out can be lucky; this is the evidence that the ranking is stable.

Forecasting caveats. The window contains an exogenous structural break (2020), so a model validated only inside that period is scoring a regime that may not repeat. Intake policy changes — a new app, a changed call flow — would invalidate the seasonal structure entirely. These forecasts describe expected call volume under normal conditions; they are not service-level commitments.

8. Anomaly detection

Weekly volume is screened with Isolation Forest (5% contamination, 500 trees) on seasonal-normalised features, so the model flags weeks that differ from their own historic norm rather than weeks that are merely busy. A transparent seasonal rule is computed alongside it as a sanity benchmark.

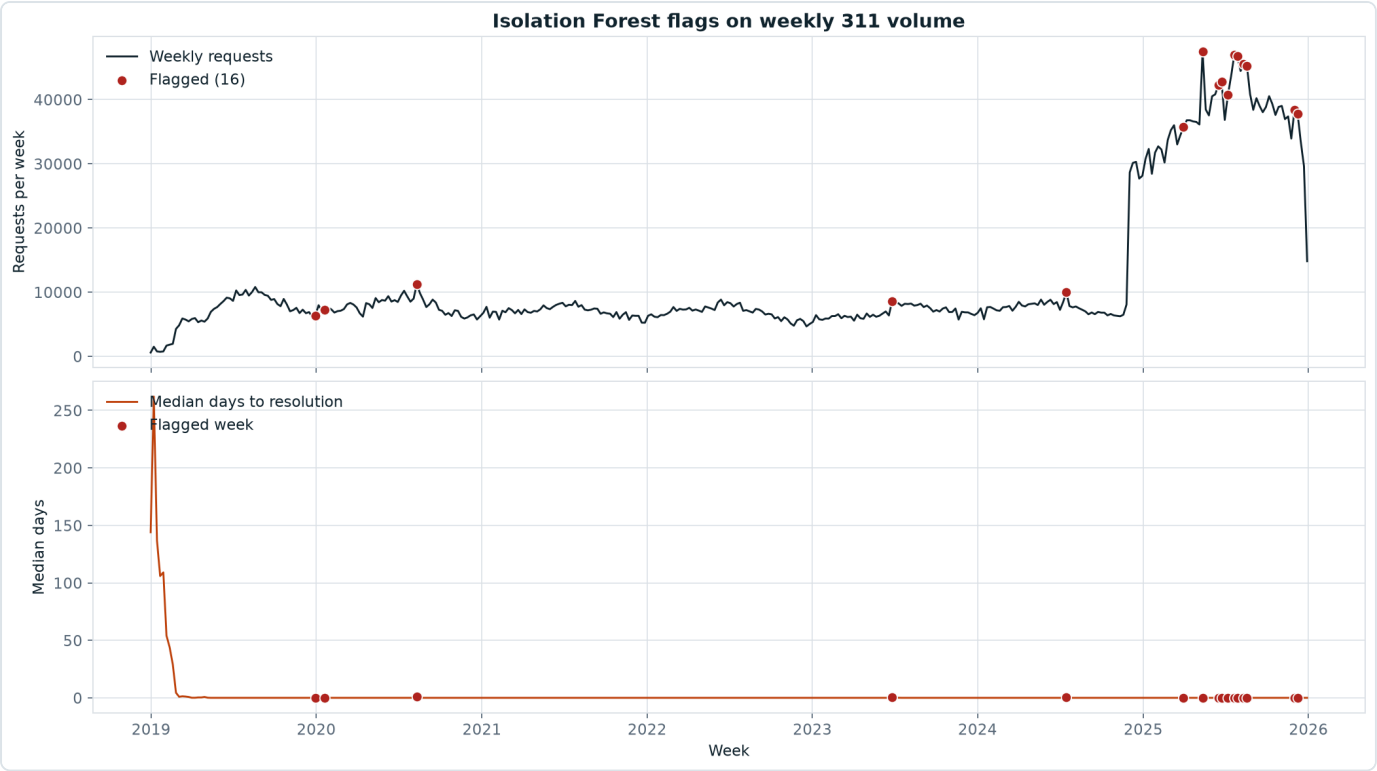


Figure 10. Weeks flagged as anomalous, against weekly volume and median resolution time.

Finding 5. The Isolation Forest flagged **16 weeks** as anomalous. The model catches weeks that a simple volume threshold misses — weeks with an abnormal severity mix or an abnormal resolution time at normal volume. Each flag is a week for a human to check against the weather record and the intake log; a flag is not a conclusion.

Anomaly caveats. Contamination is chosen, not discovered; the ranking is the trustworthy output and the count is a convenience. Multi-week events produce consecutive flags that are one incident, not several.

9. Recommendations

For city policymakers

1. **Report response time in two tiers.** Publish median resolution time for *actionable* service requests separately from administrative filings. The current single figure is dominated by one-minute closures and understates real service times by a factor of roughly 497,154. This is a reporting change, not a service change, and it is the cheapest credibility win available.
2. **Audit the slow clusters, not the slow complaints.** The significant high–high clusters identified in Figure 7 give a concrete, bounded list of **122** census tracts — and the community areas behind them — where a service audit would be most informative. Start there rather than citywide.
3. **Set a severity-differentiated service standard.** A 30-day closure on a tree trim and on a flooded basement are not the same outcome. The severity bands provide a ready-made basis for tiered targets, and the pipeline already reports the weighted measure needed to monitor them.
4. **Use the seasonal peak for capacity, not the annual average.** Capacity planned on mean annual volume is structurally short in summer.
5. **Publish tract-level response metrics as a standing dataset.** The pipeline in this repository regenerates them end to end; a quarterly publication would make the equity conversation evidence-based instead of anecdotal.

For analysts reusing this work

1. Filter on `is_actionable` before computing any timing statistic. The single most consequential step in this analysis was noticing the one-minute closures.
2. Normalise by tract population or request volume before comparing areas, and weight regressions by resolved volume.
3. Treat the 20% sample and the full-detail months as distinct scopes, and validate the sample against exact counts whenever the API is reachable.
4. Prefer median and P90 over mean for resolution time. The distribution is extremely right-skewed.

10. Reproducing this analysis

```
conda create -n chicago311 -c conda-forge python=3.12 \
    duckdb lightgbm geopandas pyarrow jupyter statsmodels \
    scikit-learn matplotlib seaborn scipy
conda activate chicago311
pip install -r requirements.txt

python src/fetch.py --all          # collect all upstream sources
python src/etl.py                  # build the DuckDB warehouse + quality report
python src/generate_notebooks.py   # build and execute the 4 notebooks
python src/build_hyper.py          # Tableau .hyper extract + tract GeoJSON
python src/build_powerbi.py        # Power BI PBIP project
python src/build_html_dashboard.py # self-contained HTML dashboard
python src/build_report.py         # this report
```

Runtime depends on the collection step, which downloads roughly 21M rows of source data (~5.8 GB) from a rate-limited public API. Everything downstream of collection runs in minutes. All random processes are seeded; all command-line entry points are idempotent.

> **Statement of limitations.** (1) The analysis is observational and the demographic associations are correlates, not causes. (2) Neither crew routing, departmental budgets, staffing nor intake channel are observed, so no mechanism is identified. (3) Closure time records an administrative act, not the date work was completed. (4) HOLC grades cover only the mapped portion of the city. (5) The redlining overlay is descriptive historical context; no causal claim is made about the 1930s maps. (6) One-third of the earliest months are a 20% sample, validated against exact counts but not identical to them.

All statistics in this report were produced by the pipeline in this repository and read programmatically from `data/processed/findings.json` and the DuckDB warehouse. No figure in the text was transcribed by hand.